EL Calls

Developer Guide — Architecture, Data Model, Edge Functions, Performance and Security

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1. Stack Overview

Frontend: React 18, Vite 5, TypeScript 5, Tailwind v3, shadcn/ui, TanStack Query, React Router v6. Backend: Lovable Cloud (managed Supabase) — Postgres 15 with RLS, Deno edge functions, Realtime broadcast, Storage. Voice: Twilio Programmable Voice + browser WebRTC. AI: Lovable AI Gateway (Gemini 2.5 Flash for chat/officer-chat, Whisper-equivalent for transcription).

1.1 Repository layout

```
src/
components/ shadcn-based UI, feature panels under dashboard/
hooks/ useAuth, useUserRole, useWebRTCCall, useAgentPresence...
integrations/supabase/ generated client + types (DO NOT EDIT)
pages/ route components
lib/ phone, normalizers, utilities
supabase/
functions/ Deno edge functions (one folder = one function)
migrations/ timestamped SQL files
load-tests/ k6 scripts and reports
public/docs/ User and Developer PDFs (this file)
```

2. Routing and Navigation

React Router v6 with a ScrollToTop component remounting on every pathname change. AdminRoute and ProtectedRoute wrap restricted routes and call useUserRole, which queries user_roles via the has_role RPC. Sidebar items are filtered by role on the client; server enforcement is the source of truth.

3. Data Model

3.1 Core tables

Table	Purpose
profiles	Public user info, linked to auth.users.
user_roles	Role assignment (admin/supervisor/agent). Separate table to prevent escalation.
clients	Customer master records with tags.

Table	Purpose
call_campaigns / campaign_types	Campaign definitions and templates.
campaign_runs / campaign_jobs	Per-execution and per-target row.
outbound_calls	One row per dialed call. 6 composite indexes added in 2026-06 migration.
outbound_call_events (+monthly partitions)	Append-only event log from Twilio status callbacks.
call_queue / call_transfers	Live routing primitives.
agent_status / agent_skills / agent_shifts	Presence, capabilities, rota.
escalation_settings / emergency_alerts	Escalation rules and raised alerts.
call_transcriptions / campaign_recordings	ASR output and recording metadata.
ivr_menu_options / supported_languages	IVR config and localisation.
audit_log / error_events / system_health_metrics	Observability.

3.2 Row Level Security

Every public table is RLS-enabled. The default pattern: enable RLS, GRANT to the appropriate roles, then create policies that delegate to security-definer helpers.

```
create or replace function public.has_role(_uid uuid, _role app_role)
returns boolean language sql stable security definer
set search_path = public as $$
select exists(select 1 from public.user_roles where user_id=_uid and role=_role)
$$;

create policy "Admins manage all" on public.call_campaigns
for all to authenticated using (public.has_role(auth.uid(),'admin'));
```

Partitioned tables (outbound_call_events_YYYY_MM) inherit RLS from the parent and are explicitly RLS-enabled on creation by create_outbound_call_events_partition().

4. Edge Functions

All functions live under *supabase/functions/<name>*. Naming convention: kebab-case folder; index.ts as entrypoint. Shared utilities in *_shared/* include twilio-verify (signature validation), idempotency, rate-limit, audit, phone and monitor.

Function	Auth	Purpose
ai-voice-call	Twilio signature	Initial TwiML for outbound dialer.
ai-voice-call-status	Twilio signature	Inserts every Twilio status into outbound_call_events (pure append, no UPDATE).
ai-voice-call-dtmf*	Twilio signature	DTMF gather flows including appointment + language.
ai-voice-call-bridge-ivr	Twilio signature	Connects callers to the configured IVR menu.
process-call-events	CRON_SECRET	Drains outbound_call_events into outbound_calls in batches of 500.
campaign-scheduler	Staff JWT	Plans which campaigns to run within dialing windows.
campaign-enqueue	Staff JWT	Enqueues campaign_jobs into call_queue.
campaign-worker	CRON_SECRET	Pops jobs and triggers Twilio dials.
campaign-control	Staff JWT	Start/pause/resume campaigns.
retry-campaign-calls	CRON_SECRET or staff	Re-attempts failed jobs per retry policy.
reconcile-call-status	CRON_SECRET	Hourly reconciliation against Twilio API.
smart-call-routing	Staff JWT	Scores agents by skills+proficiency+performance.
analyze-call-sentiment	Internal	Lovable AI sentiment per transcript turn.
transcribe-call / voice-to-text	Staff JWT, 10MB cap	Whisper transcription.
officer-chat / ai-policy-notes	Authenticated	Gemini 2.5 Flash assistants grounded in knowledge_base.
send-agent-invitation	Staff JWT	Server forces invitedBy = caller identity.
send-escalation-notification	Staff JWT	Pulls recipients from escalation_settings only.
send-appointment-email / send-payment-email	Internal	Transactional emails via SendGrid.
manage-agents / manage-app-secrets	Admin JWT	Admin CRUD.
twilio-recent-calls / twilio-verify	Staff JWT	Diagnostics.

Function	Auth	Purpose
suggest-tag	Staff JWT	Heuristic + AI tag suggestions.
sample-data	Staff JWT	Seed demo data.
gdpr-export / gdpr-delete	Admin JWT	Privacy operations.
realtime-chat	Authenticated	WebSocket relay for live chat panel.

4.1 Error handling contract

Edge functions log full errors server-side and return a generic message to the client to avoid information disclosure:

```
try { /* work */ }
catch (e) {
console.error('fn-name failed', e);
return new Response(JSON.stringify({ error: 'An unexpected error occurred' }),
{ status: 500, headers: corsJson });
}
```

4.2 Twilio webhook security

Inbound Twilio callbacks (status, DTMF, IVR bridge) validate the X-Twilio-Signature header via *_shared/twilio-verify.ts* using the auth-token secret. Replay protection uses the idempotency helper keyed on CallSid+CallStatus.

5. Performance Architecture

5.1 Index strategy

Migration 2026-06-22 added six composite indexes on outbound_calls:

- twilio_call_sid (webhook lookups, O(1))
- campaign_id + created_at DESC (campaign dashboards)
- call_status + created_at DESC (live queues)
- agent_email + created_at DESC (agent performance)
- phone_number (fallback matching)
- client_id + created_at DESC (client-level queries)

5.2 Event-log + partitioning

ai-voice-call-status is hot during campaigns: 4-6 callbacks per call. The previous design did an UPDATE on outbound_calls per callback and contended on the same row. The new design:

- Webhook does pure INSERT into outbound_call_events (range-partitioned by month).
- process-call-events runs every 5-10s on a cron, batching 500 rows.
- It resolves call IDs via twilio_call_sid or parent_sid index, then applies the final state with a single UPDATE per call inside one transaction.
- create_outbound_call_events_partition() pre-creates next month's partition and enables RLS automatically.

```
-- drainer (simplified)
select public.process_outbound_call_events(500);
```

5.3 Load test results (k6)

Scenario	Load	p95	Throughput	Notes
Campaign execution	200 RPS, 5m	220 ms	98.4% OK	DB CPU 55%.
Concurrent dialing	500 active calls	n/a	Stable	Twilio rate is the cap.
Webhook processing	1200 RPS	85 ms	100% OK (post-fix)	Was 22% errors before event-log.
Dashboard traffic	200 vUsers, 10m	180 ms	99.9% OK	Indexes cut p95 by 6x.

Max sustained capacity: ~130k calls/day per project on Lovable Cloud Pro plan; bottleneck shifts to Twilio concurrent-call quota above that.

6. Security

- Roles stored in user_roles table only; checked via has_role security-definer.

- RLS on every public table and every partition. GRANTs explicit, never relying on defaults.
- Edge functions classify auth: public + Twilio-signed, authenticated, staff JWT, admin JWT, CRON_SECRET.
- No SUPABASE_SERVICE_ROLE_KEY usage on the client; only edge functions read it.
- Generic error messages on all public surfaces; full errors logged via console.error.
- Rate-limiting helper in _shared/rate-limit.ts protects expensive endpoints.
- Audit-log entries written for all role changes, GDPR ops, secret reads, escalations.
- Auth redirects pinned to https://elcalls.lovable.app to avoid open-redirect.
- Sign-out forces redirect to '/' and clears all client state.

7. Realtime and Presence

Supabase Realtime broadcast is used for: live transcription updates, queue and transfer changes, supervisor live view, and officer-chat streaming. useAgentPresence sends a heartbeat every 30s and flips users to *away* after 5 minutes of inactivity. WebRTC signalling rides the same Realtime channel.

8. WebRTC Calling

useWebRTCCall sets up RTCPeerConnection per call, with ICE servers fetched from a signed edge function. SDP offers/answers and ICE candidates broadcast through a Supabase Realtime channel scoped to the call ID. The LiveKit credentials slot exists for a future migration to LiveKit SFU when concurrent browser-to-browser calls exceed the mesh limit.

9. AI Integrations

- Officer Chat and AI Policy Notes use Lovable AI Gateway with google/gemini-2.5-flash by default.
- Tag suggestions use a heuristic pass plus an AI fallback.
- Sentiment runs per transcript turn and writes to call_transcriptions.sentiment.
- Transcription uses Whisper via voice-to-text/transcribe-call; 10MB payload cap.

All AI calls go through the gateway — no third-party API keys are required from the customer for chat/embeddings/STT/TTS unless they want a specific provider.

10. Local Development

```
bun install
bun run dev # vite on :8080
bunx vitest run # unit tests
bunx tsgo --noEmit # typecheck (preferred over tsc)

# edge functions: deploy via Lovable Cloud, logs in dashboard
# never edit src/integrations/supabase/{client,types}.ts (auto-gen)
```

11. Migrations

Every CREATE TABLE in *public* is followed in the same migration by GRANTs and RLS enable + policies. The repo includes a migration that backfills RLS on existing partitions and adds the create_outbound_call_events_partition() helper.

12. Observability

- error_events captures handled exceptions from edge functions.
- system_health_metrics records throughput, queue depth, ASR latency.
- audit_log records every sensitive action (role changes, secret reads, GDPR ops).
- Lovable Cloud function logs are the primary surface for live debugging.

13. Testing

- Unit tests: vitest, including src/lib/phone.test.ts and supabase/functions/_shared/*.test.ts.
- Load tests: k6 scripts under load-tests/scripts (01 campaign, 02 dialing, 03 webhook, 04 dashboard).
- Run all: bunx vitest run; load report at load-tests/reports/REPORT.md.

14. Deployment

Use the Publish action in Lovable. Custom domains map to the published URL. Cron schedules for process-call-events, campaign-worker, reconcile-call-status and retry-campaign-calls are configured in Lovable Cloud. CRON_SECRET protects all cron-triggered endpoints.

15. Extending the Platform

- New campaign type: insert into campaign_types, add any IVR menu options, deploy.
- New language: insert supported_languages row, upload audio via LanguageAudioUploader, add translation strings.
- New role: extend app_role enum, add policies, update AdminRoute logic.
- New AI feature: prefer Lovable AI Gateway; never embed third-party keys in client.

16. Coding Conventions

- Tailwind tokens only; no hardcoded colour utilities (no text-white, bg-[#hex]).
- Long forms broken into tabs so action buttons stay visible.
- Wow aesthetic: gradients, glassmorphism, micro-interactions (ripples, confetti).
- Officer chat always rendered as a floating non-modal right-side panel.
- Sidebar collapsible from the left; scroll to top on every nav.

17. Appendix: Useful RPCs

RPC	Returns	Used for
has_role(uid, role)	boolean	RLS policies
is_admin() / is_staff() / is_supervisor_or_admin()	boolean	Edge function gating
process_outbound_call_events(limit)	integer	Drainer cron
create_outbound_call_events_partition(date)	void	Monthly partition creation

18. Active Directory (SAML SSO) Setup

The Active Directory configurator (Setup !’ People !’ Active Directory) provides a guided, auto-validated workflow for connecting EL Calls to a corporate IdP. Implementation: src/components/dashboard/ActiveDirectorySetup.tsx.

18.1 Supported IdP Types

A discriminated idpTemplates map drives the dynamic field set per directory type: Microsoft Entra ID (Azure AD), AD FS, Okta, OneLogin, Google Workspace and Generic SAML 2.0. Selecting a type swaps the required fields and resets verification state.

18.2 Draft Save & Resume

Configuration is persisted as a draft in localStorage under key el:ad-setup:draft:v1. The draft preserves: selected IdP type, captured field values, allowed domains, pilot users, default role, enforcement toggles, break-glass email, and the verified metadata snapshot (when status is 'ok'). On dialog open the draft is hydrated and the user is notified. A 'Draft saved' badge plus a 'Resume Active Directory setup' CTA appear on the card when a draft exists. Submission clears the draft via clearDraft().

18.3 Entra ID Consistency Validation

When the IdP type is entra, an entraIssues memo runs the following checks before any checklist item turns green:

- Tenant ID matches GUID regex ^[0-9a-f]{8}-...-[0-9a-f]{12}$
- Enterprise App Object ID matches the same GUID regex
- Federation Metadata URL host is login.microsoftonline.com (or .us)
- Metadata URL contains the supplied Tenant ID
- Metadata URL path includes /federationmetadata/
- After fetch, the returned entityID references the Tenant ID

18.4 Test SSO Button

runSsoTest() performs a live probe against the IdP. Preconditions: all required fields supplied, metadata verified, and (for Entra) consistency checks passing. The probe fetches the SSO endpoint extracted from the verified metadata using mode: 'no-cors'; an opaque response counts as reachable. The result is stored in ssoTest state as idle | running | success | error, surfaced both inline and as a checklist row with a specific failure reason.

18.5 Auto-Verified Production Checklist

The checklist is computed by an autoChecks memo — users cannot tick items manually. Each entry exposes label, done, and a why string explaining the current state. Green rows show a confirmation; red rows show the precise blocker.

- All required IdP fields supplied — lists missing field labels when red.
- IdP details consistent — surfaces the first Entra consistency error.
- IdP metadata verified & reachable — reports HTTP/parse failure or asks for Verify click.
- Live SSO test passed — shows the probed hostname or the specific failure reason.
- Email domains allowlisted — lists malformed domains.
- Pilot users supplied (2-5 valid emails) — explains count or format violation.
- Single Logout + SSO enforcement enabled.
- Break-glass admin email captured (or explicitly disabled).

Submit is enabled only when every check is green. On submit the component dispatches a lovable:saml-sso-submit CustomEvent with the full payload and clears the draft. Server-side SAML trust finalisation is handled by Lovable Cloud SSO configuration.

18.6 Service Provider Constants

- Entity ID: https://elcalls.lovable.app/saml/metadata
- ACS / Reply URL: https://elcalls.lovable.app/saml/acs
- Single Logout URL: https://elcalls.lovable.app/saml/slo
- NameID format: emailAddress